Ben Schweitzer

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For the past 11 years, I've been enjoying an incredible career in the design, production, and deployment of energy storage and conversion systems. Starting from the fundamental sciences involved in Li-ion battery degradation and thermal runway phenomenon, I've been able to apply that knowledge to design high performance, high reliability Li-ion battery systems across both ground and aerial applications. As I've grown professionally, that knowledge has allowed me to guide and influence the preliminary design and optimization of larger and more complex energy systems; most recently, Amazon's electric delivery fleet of 100,000 Rivian Electric Delivery Vehicles (EDV) and L2/L3 charging infrastructure!

Work Experience

Amazon – Seattle, WA

Principal Engineer, Electrification – Last Mile Global Fleet and Products (April 2022 – Current)

Principal Systems Engineer leading a research science group responsible for architectural decisions in the electrification of Amazon's Last Mile delivery network.

- Developed mathematical models to simulate fleet charging operations and optimize charging hardware and control strategies to minimize CapEx and charging electricity costs in operations.
- Led the introduction of battery life modeling into our EV deployment simulation infrastructure, allowing for trade studies between charge rates and battery size with respect to fleet lifetime and expected CapEx.

Manager, Battery Systems Engineering – Last Mile Global Fleet and Products (March 2020 – April 2022)

Senior Engineer/Manager responsible for energy systems requirements on pilot electric vehicles (EVs), as well as the Rivian Electric Delivery Vehicle (EDV).

- Worked with Rivian to design, simulate, and develop a next generation energy storage system (ESS) for the EDV, reducing cost while meeting Amazon energy requirements for our delivery routes.
- Developed physics-based models for delivery route energy estimation, achieving a MAPE of 13% across 3 vehicle variants without machine learning techniques. Models are used by internal partner teams to plan EV deployments 3 years out.

Manager, Fuels and Energy Conversion – Amazon Prime Air (April 2019 – March 2020)

Hired and developed a team of 8 hardware and software engineers responsible for all on-vehicle energy storage and conversion systems, along with on-ground charging hardware.

- The team lead the design, build, and validation activities for 3 generations of Prime Air vehicle ESS, across multiple Li-ion cell format (pouch, cylindrical). Achieved pack level energy density of 185 Wh/kg while meeting 4C continuous/ 8C peak discharge rates with passive cooling.
- Scaled battery pack production for latest drone design to hundreds of units per year with a 3rd party contract manufacturer.
- Designed and delivered 65 bespoke charging systems for the ESS, capable of remote deployment to drone test ranges.

Sr. Battery Systems Engineer – Amazon Prime Air (May 2016 – April 2019)

As the first battery engineer hire on the team, I led the definition and requirements development for Prime Air's first bespoke battery system. I was also the initial build engineer (and technician!) for the in-house assembly of the pack.

- Defined and procured cell and pack test equipment for Li-ion battery testing (36 cell channels/16 pack channels)
- Ran all cell characterization testing, and developed 2 RC pair electro-thermal battery models for use in Prime Air's MDAO aircraft design tool.
- Incubated and developed business relationship with multiple Tier 1 battery cell OEMs, allowing us to source cells directly for evaluation and LRIP.

AllCell Technologies – Chicago, IL

Cell and Module Engineer, Research and Development (June 2012 – October 2015)

- Responsible for Li-ion cell and module electrical and thermal testing/characterization.

- Developed cell and module electrochemical models, including equivalent circuit as well as single particle physics approaches.
- Design and executed thermal runaway tests across cell, module, and packs.
- Investigated the efficacy of multiple types of phase change composite materials on thermal runaway propagation.
- Developed long term cell aging models, capable of consuming customer usage profiles and predicting capacity loss and resistance growth over time and energy throughput.
- Lead engineer for a clad metal current collector development program, including projection spot welding process development; reduced current collector ohmic losses by 65% in stationary energy storage product.

General Motors – Warren, MI

Test Engineer, Global Battery Systems Lab (November 2011 – June 2012)

- Responsible for Volt Gen 2 cell and pack test setup and execution activities

Publications and Awarded Patents

Publications

M. Salameh, S. Wilke, B. Schweitzer, P. Sveum, S. Al-Hallaj, *Thermal State of Charge Estimation in Phase Change Composites for Passively Cooled Lithium Ion Battery Packs,* IEEE Transactions on Industry Applications, Volume 54, 2017, 426-436.

S. Wilke, B. Schweitzer, S. Khateeb, S. Al-Hallaj, *Preventing thermal runaway propagation in lithium ion battery packs using a phase change composite material: An experimental study,* Journal of Power Sources, Volume 340, 2017, 51-59.

F. A. LeBel, S. Wilke, B. Schweitzer, M. A Roux, S. Al-Hallaj, J.P.F Trovao, *A Lithium-Ion Electro-Thermal Model of Parallelized Cells*, IEEE 84th Vehicular Technology Conference, Sept 2016.

S. Wilke, B. Schweitzer, S. Al-Hallaj, *Semi-Empirical Modeling of Capacity Fade: A Practical Approach for Battery Pack Manufacturers,* ECS Transactions, Jun 20, 2016.

B. Schweitzer, S. Wilke, S. Khateeb, S. Al-Hallaj, *Experimental validation of a 0-D numerical model for phase change thermal management systems in lithium-ion batteries*, Journal of Power Sources, Volume 287, 2015, 211-219.

P. Taheri, A. Mansouri, B. Schweitzer, M. Yazdanpour, M. Bahrami*, Electrical Constriction Resistance in Current Collectors of Large-Scale Lithium-Ion Batteries*, Journal of the Electrochemistry Society, Vol. 160, 10, 2013, A1731 - A1740.

Awarded Patents

Charging Mat for Unmanned Aircraft – US 10,418,830 B1
Mechanically Controlled Pre-Charge System – US 10,193,370 B1
Power Supply Monitoring System Using Optical Estimation – US 10,393,821
Power Supply Monitoring System Using Strain Gages – US 10,809,309
Power Supply Monitoring Systems and Methods Using Ultrasonic Sensors – US 11,175,346
Aerial Vehicle Fleet Maintenance Systems and Methods – US 11,374,415

Education

M. Eng. Energy Systems | University of Michigan, Ann Arbor 2011
B.S. Mechanical Engineering | University of Michigan, Ann Arbor 2004